BRAC UNIVERSITY

Principles of Physics-II (PHY-112)

Department of Mathematics and Natural Sciences

Assignment: 01 — **Section**: 03

Dispatch Date: February 26, 2025

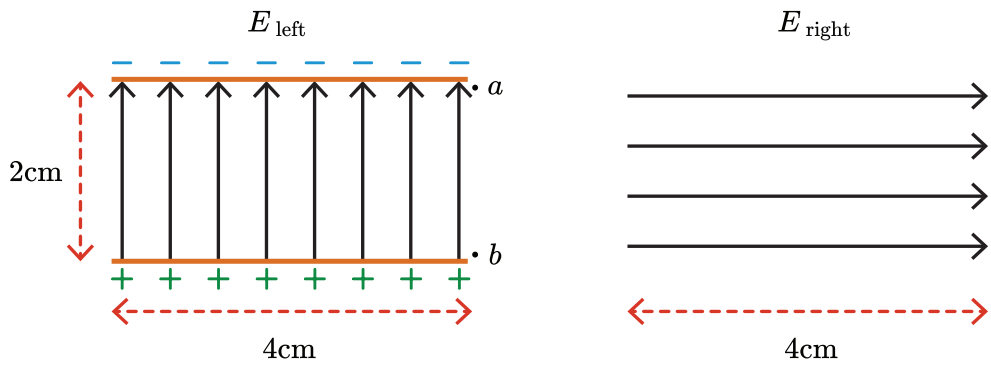
Submission Deadline: March 6, 2025

Duration: 6 Days | **Spring 2025 (10F-29C)** | Marks: 30

Assignment Instructions

- Write your assignment on A4-sized paper. Use SI units. Properly staple and organize your pages.
- Attempt the questions set for your designated section only. Follow the deadline as written above.
- Ensure your solutions are clear, concise, and well-organized. Add diagrams where needed. Expand your answers when necessary. Do not spit out equations with no context. Steps hold marks.
- **Plagiarism, cheating, copying someone else's script, or the blatant use of AI tools** will result in a 50% decrement or a zero score in extreme cases.
- Late submissions will be penalized at a rate of **5% per day** for three days. No submissions will be accepted after that. Especially after the solution is posted.

1. A uniform electric field, spanning 4 cm horizontally is directed Eastward to your right. This field is assigned to accelerate (from rest, in a direction that speeds up the beam) a beam of electrons, giving them a uniform velocity as they exit the region. The electrons then enter, perpendicularly, a second uniform electric field of magnitude $E_{\text{left}} = 4\,\text{MN}\,\text{C}^{-1}$ directed Northward to your left. The parallel electrodes in the leftward region are separated by 2 cm and extend 4 cm in length. The electron beam enters the left field from point a.



(a) Determine the magnitude of the electric field E_{right} on the right required to accelerate the electron beam to 90% of the speed of light over the distance of the field range of 4 cm. Additionally, determine the direction in which the beam exits this region. **(3)**

(b) Upon entering the leftward field with the velocity obtained by the acceleration found in (a), compute the time interval required for the beam to traverse the parallel plates. **(3)**

(c) If the electron beam successfully clears the plates (passes the field region without hitting either plate), determine the velocity vector and speed of the beam at the instant it emerges/exits from the field. Calculate the angle the particle's trajectory makes with the positive x-axis at this instance, measured counterclockwise. **(4)**

(d) Suppose the electron beam is replaced with a proton beam entering the field from point b (E_{right} is not accelerating it.). Would the time interval found in (b) be sufficient for the protons to clear the plates? [Hint: Calculate vertical displacement using the proton's acceleration.] **(5)**

2. An electric dipole with a dipole moment of magnitude p is placed at various orientations in an electric field $\vec{E}$ that is directed to the left. (a) What orientation of the dipole will result in maximum torque directed into the page? What, then, is the electric potential energy? (b) What orientation of the dipole will give zero torque and maximum electric potential energy? What type of equilibrium is this: stable or unstable? Drawing a diagram in your answer script is mandatory as a visual guide behind your explanation. **(5)**

3. You have four charged particles to form a square, each at a corner, with side length $a = 5.00\,\text{cm}$. The charges are $q_1 = 10.0\,\text{nC}$ (upper-left), $q_2 = -20.0\,\text{nC}$ (upper-right), $q_3 = 20.0\,\text{nC}$ (lower-right), and $q_4 = -10.0\,\text{nC}$ (lower-left). Calculate the net electric field at the center of the square in **unit-vector notation**. **(4)**

4. Find (i) what equal positive or negative charge must be placed on both the Earth and the Moon to neutralize their mutual gravitational attraction. Why is the distance between the Earth and the Moon irrelevant to this calculation? (ii) How many kilograms of Hydrogen ions (protons) would be needed to generate this charge? **(3)**

5. Two charges, $+2q$ (at $x = 5\,\text{cm}$) and $+5q$ (at $x = -5\,\text{cm}$) are placed $10.00\,\text{cm}$ apart on the x-axis. To ensure balance, a third particle with charge q_3 must be positioned in the plane (other than at infinity) to *stabilize* the system. Find the location of the third charged particle in the component form. You must start by checking at which interval the equilibrium must be: $x < -5\,\text{cm}$ or $-5\,\text{cm} < x < 5\,\text{cm}$ or $x > 5\,\text{cm}$. **(3)**

Important Constants and Order of Magnitude Prefixes	
Constants	**Prefixes**
Vacuum Permittivity, ϵ_0 $8.854 \times 10^{-12}\,\text{F}\,\text{m}^{-1}$	10^{-15} = f (femto) 10^{-12} = p (pico)
Vacuum Permeability, μ_0 $4\pi \times 10^{-7}\,\text{T}\,\text{m}\,\text{A}^{-1}$	10^{-9} = n (nano) 10^{-6} = μ (micro)
Speed of Light, c $3 \times 10^{8}\,\text{m}\,\text{s}^{-1}$	10^{-3} = m (milli) 10^{-2} = m (centi)
Electron Charge, e $1.602 \times 10^{-19}\,\text{C}$	10^{2} = h (hecto) 10^{3} = k (kilo)
Proton mass, m_p $1.673 \times 10^{-27}\,\text{kg}$	10^{6} = M (mega) 10^{9} = G (giga)
Electron mass, m_e $9.109 \times 10^{-31}\,\text{kg}$	10^{12} = T (tera) 10^{15} = P (peta)

May the Force be with you!